

Introduction to Digital Trace Data: Quality, ethics, and analysis

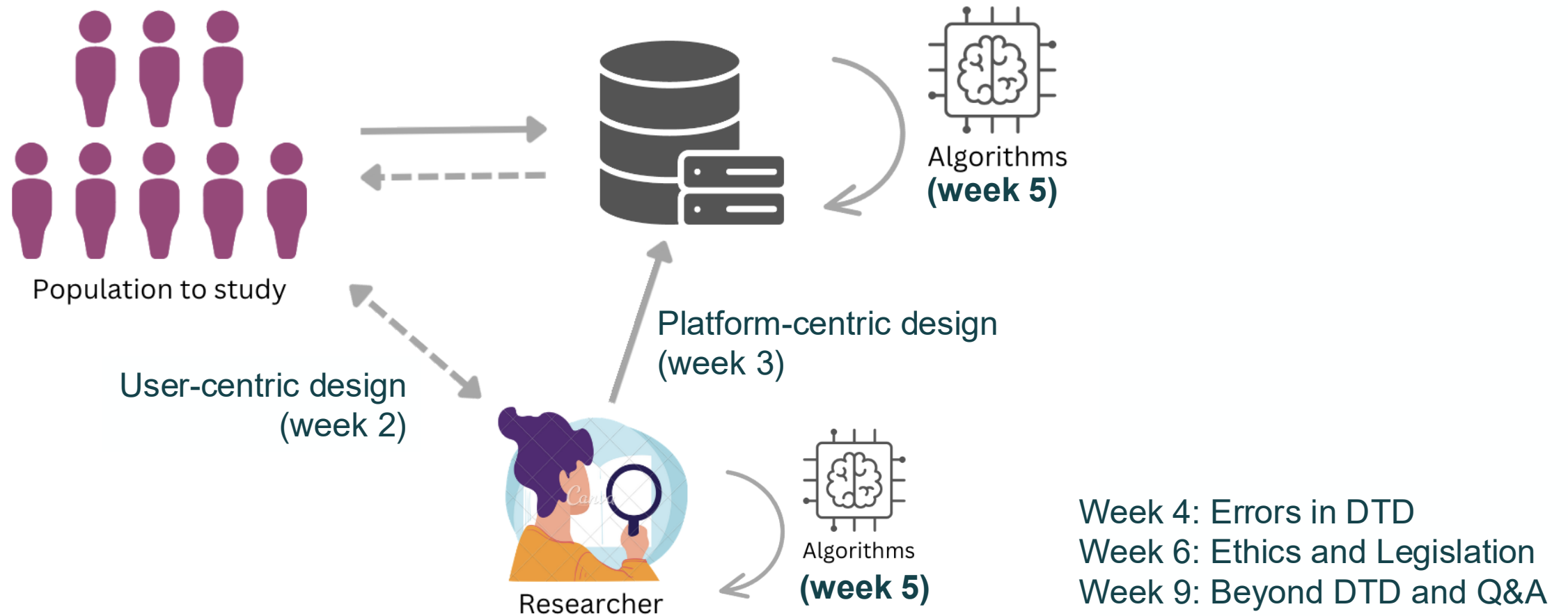
Lecture 5: The role of AI in DTD

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Where are we?



TODAY

Lecture

1. What is machine learning and how can we use it?
 - Explain machine learning in your own words
2. Why are ML models biased? Five sources of bias
 - Distinguish the five sources of bias in an ML model and give an example of each
3. How ML affects digital trace data and societies
 - Explain how ML changes what we measure in DTD, and how it can reinforce power
4. Measuring and dealing with bias in ML
 - Compute and compare error rates across groups to assess bias

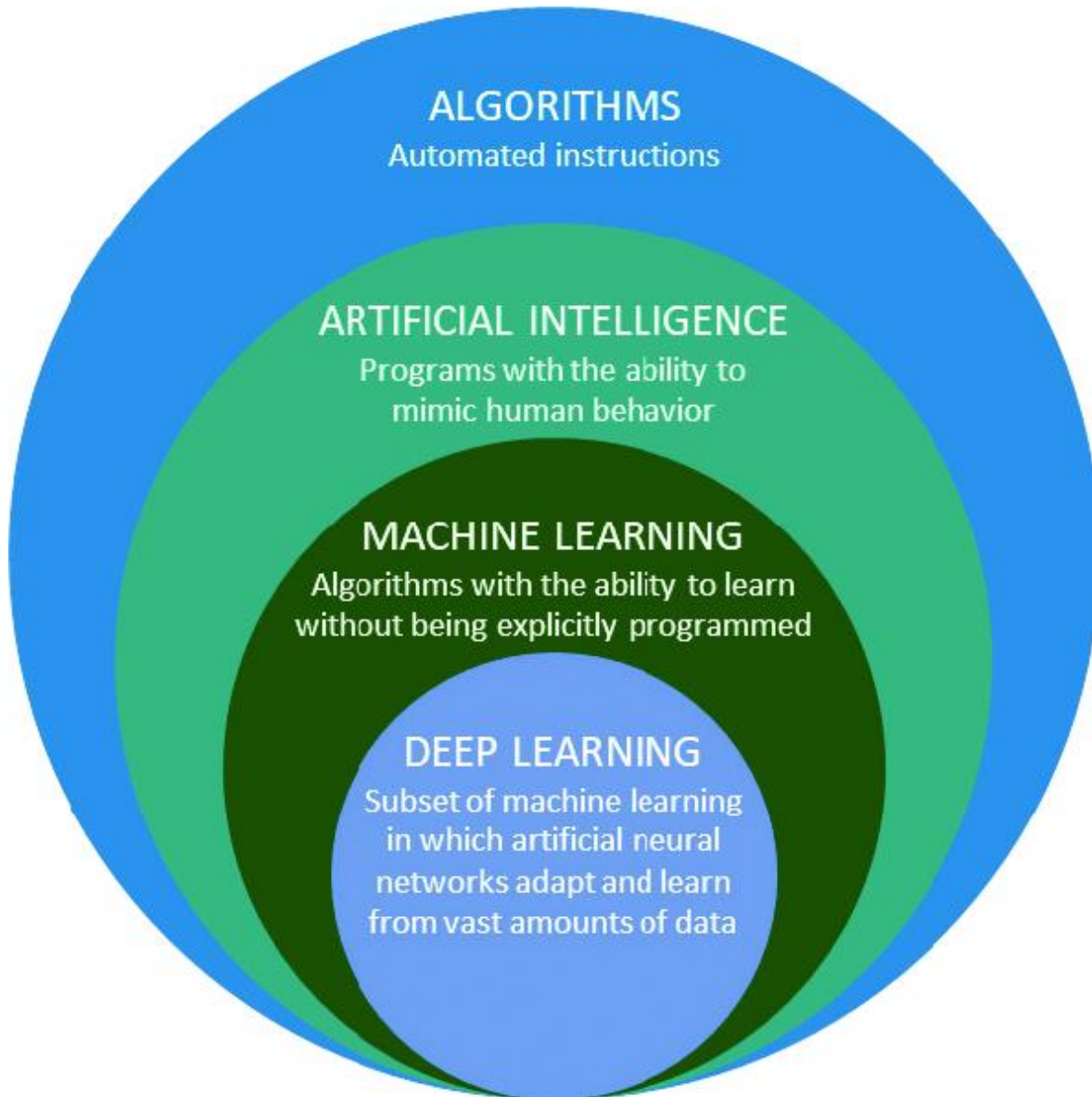
Lab

Discuss power dynamics behind ML models

Apply an ML model to text data

Audit an ML model

1. What is machine learning and how can we use it?



Machine Learning

"A computer program is said to learn from **experience** E with respect to some class of **tasks** T and **performance measure** P if its performance at task T , as measured by P , improves with experience E ." (Samuel/Mitchell, 1959/1997)

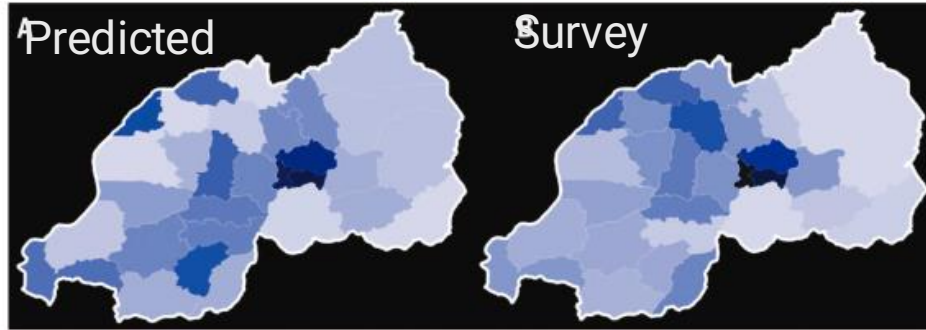
- **Experience:** Data (e.g. comments from TikTok)
- **Task:** Goal of the model (e.g. predict hate speech)
- **Performance measure:** Accuracy, R^2 , etc

Supervised ML: the model learns from labelled examples ($X \rightarrow Y$) and predicts Y for new cases.

Is a linear regression a machine learning model?

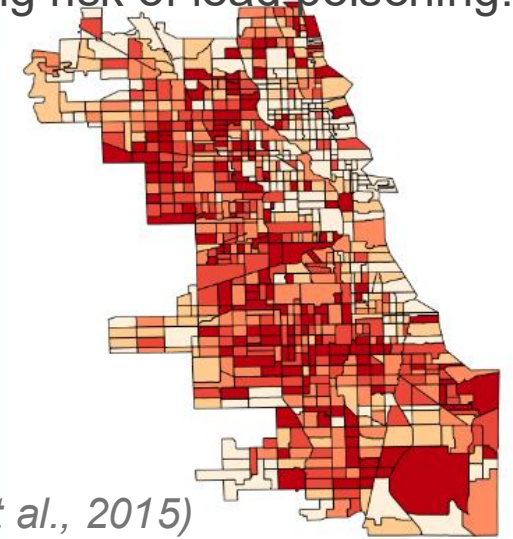
Using ML: prediction (supervised ML)

Predicting wealth/SES:

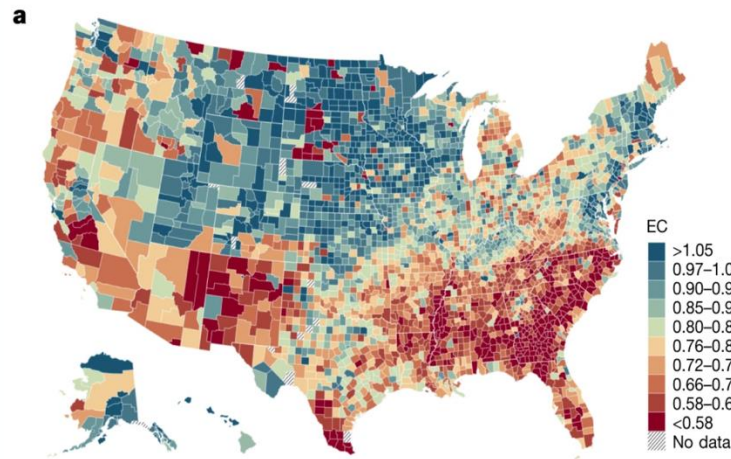


(Blumenstock et al., 2015)

Predicting risk of lead poisoning:

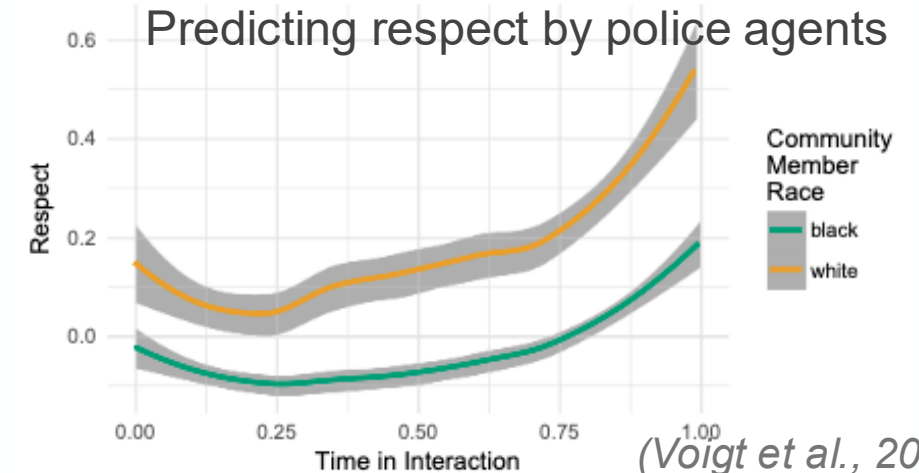


(Potash et al., 2015)



(Chetty et al., 2022)

Predicting respect by police agents



(Voigt et al., 2017)

2. Why are ML models often biased/unfair*?

*The model performance is different for different subgroups

Errors in algorithms

We need to consider the errors of every algorithm:

- How often they fail?
- For whom do they fail? (***bias/fairness***)

No model will be perfect, but we need to understand when and for whom they fail.

How often do they fail? The confusion matrix

	Predicted criminal	Predicted not criminal
Criminal	True positive	False negative
Not criminal	False positive	True negative

For whom do they fail?

Group A	Predicted criminal	Predicted not criminal
Criminal	10	10
Not criminal	1	100

Group B	Predicted criminal	Predicted not criminal
Criminal	10	1
Not criminal	10	100

Fitting and using ML models (supervised ML)

1) Training data

Bias here: sample · label · features



2) Model

Bias here: ML pipeline



3) New data

Bias here: application

**X: Flights
to Russia**

Y: Criminal



3



12



1



80



If $X > 10$: criminal
If $X \leq 10$: not criminal

**X: Flights
to Russia**



20



6



3

Five sources of bias: five questions to ask

1. **Sample bias:** Who is in the training data? Do they look like the people we will predict for?
2. **Label (outcome) bias:** Does Y mean the same thing for every group?
3. **Feature bias:** Does X mean the same thing for every group?
4. **ML pipeline bias:** Did our cleaning, code, model or metric create errors for some groups?
5. **Application bias:** Is the model used in the same way for every group?

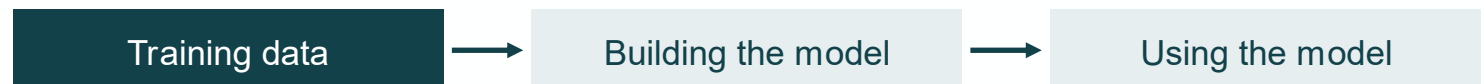
1/5 Sample bias

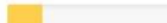
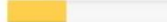
Sample bias: the training data do not represent the people the model will be used on.

X: Flights to Russia	Y: Criminal
 3	
 12	
 1	
 80	

Your sample may be mostly Dutch people, who may not go to Russia often unless they are criminals or have business. This could happen because:

- you are using data tracking only Dutch tourists
- you collected data during COVID times, or 10 years ago (drift)



Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0% 	79.2% 	100% 	98.3% 	20.8% 
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 	33.8% 
 IBM	88.0% 	65.3% 	99.7% 	92.9% 	34.4% 



Sample bias in practice: Gender Shades. Gender classifiers from Microsoft, Face++ and IBM: errors up to about 35% for darker-skinned women, under 1% for lighter-skinned men. Widely used face datasets were up to 80% white and 75% male (Buolamwini & Gebru, 2018; Big Data and Social Science 11.6.2).

2/5 Label (outcome) bias

Label bias: the label means something different for some groups.

X: Flights to Russia	Y: Criminal
 3	
 12	
 1	
 80	

If police mostly look for criminals among foreigners, the label “non-criminal” for Dutch people means something else (lack of policing, not lack of crime).

We can always expect label bias if the humans creating the labels are themselves biased.

Training data



Building the model



Using the model

Dissecting racial bias in an algorithm used to manage the health of populations

ZIAD OBERMEYER , BRIAN POWERS, CHRISTINE VOGELI, AND SENDHIL MULLAINATHAN  [Authors Info & Affiliations](#)

SCIENCE • 25 Oct 2019 • Vol 366, Issue 6464 • pp. 447-453 • DOI: 10.1126/science.aax2342

↓ 139,001 🗨 1,266



Racial bias in health algorithms

The U.S. health care system uses commercial algorithms to guide health decisions. Obermeyer *et al.* find evidence of racial bias in one widely used algorithm, such that Black patients assigned the same level of risk by the algorithm are sicker than White patients (see the Perspective by Benjamin). The authors estimated that this racial bias reduces the number of Black patients identified for extra care by more than half. Bias occurs because the algorithm uses health costs as a proxy for health needs. Less money is spent on Black patients who have the same level of need, and the algorithm thus falsely concludes that Black patients are healthier than equally sick White patients. Reformulating the algorithm so that it no longer uses costs as a proxy for needs eliminates the racial bias in predicting who needs extra care.



Label bias in practice: health care costs used as a label for health needs. Less money is spent on Black patients with the same needs, so the algorithm judged them healthier than equally sick White patients. Obermeyer et al. (2019), Science, <https://doi.org/10.1126/science.aax2342>

Humans are biased too

Case Study: Blind Auditions & Women in Orchestras (U.S.)

Goldin & Rouse, American Economic Review (2000)

In the top five U.S. orchestras, women were <5% of players in 1970 versus ~25% by the mid-1990s.

Why? Gender bias. Most major U.S. orchestras adopted 'blind' auditions (with a screen) in the 1970s–80s.

Blind auditions increased the probability a woman advanced from prelim rounds by ~50%, and raised the chance she won the final round by severalfold.

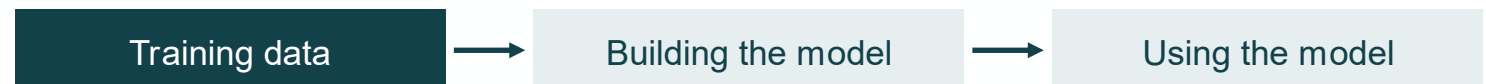
If we train a hiring model on past decisions, the label "hired" carries this bias.

3/5 Feature bias

Feature bias: a feature means something different, or is measured less well, for some groups.

	X: Flights to Russia	Y: Criminal
	3	
	12	
	1	
	80	

Flights to Russia: for Russians, flights may mean family visits.



Feature bias in practice: DUO student-grant checks

DUO (Dutch organization for student finance, loans, etc) wanted to identify students who might incorrectly be receiving the higher student grant for living away from their parents.

To select students for home visits, it used a risk score based on three apparently neutral features: Age, Type of education, Distance between the student's and parents' addresses

The Dutch Data Protection Authority (AP) concluded that this system was discriminatory and unlawful against migrants. Why?

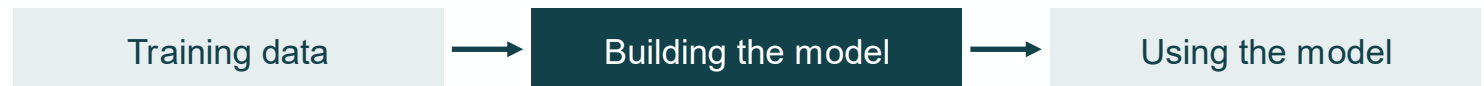
Sources: DUO press release (1 March 2024); Autoriteit Persoonsgegevens (November 2024)

4/5 ML pipeline bias

ML pipeline bias: errors we introduce when we clean, link, process or model the data.

X: Flights to Russia	Y: Criminal
 3	
 12	
 1	
 80	

- A script that cannot handle Russian names drops them in the process.
- A model or performance metric that focuses mostly on the majority class.
- Errors in your code; using the wrong model.



Pipeline bias in practice: TennCare Connect system (built by Deloitte and other contractors for more than \$400 million)

TECH NEWS

Judge Rules \$400 Million Algorithmic System Illegally Denied Thousands of People's Medicaid Benefits

Thousands of children and adults were automatically terminated from Medicaid and disability benefits programs by a computer system that was supposed to make applying for and receiving health coverage easier.

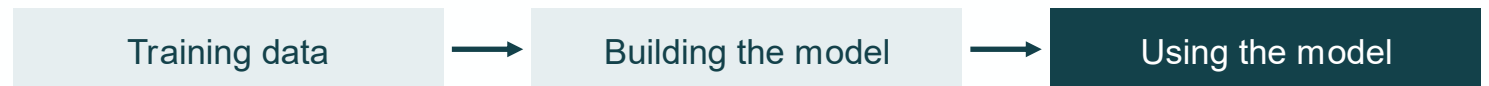
“The system often *doesn't load the appropriate data*, assigns beneficiaries to the wrong households, and makes incorrect eligibility determinations”

5/5 Application bias

Application bias: the model and data are right, but the model is applied in a biased way.

X: Flights to Russia	Y: Criminal
 3	
 12	
 1	
 80	

A manager always trusts the system when it classifies foreigners as criminals but ignores it for Dutch people.



Now all five at once: Amazon's hiring model

Amazon trained a model to rank job applicants on CVs submitted over a 10-year period. Most came from men.

- It penalized CVs that included the word "women's" (as in "women's chess club captain") and downgraded graduates women's colleges.
- Amazon edited the model to make them neutral to these particular terms. The team was eventually disbanded.
- Recruiters looked at the tool's recommendations, but according to Amazon never relied solely on them.

In pairs (3 min): which of the five sources of bias can you find?

Sample · Label · Feature · Pipeline · Application

Source: Reuters (J. Dastin, 10 October 2018); also discussed in Data Feminism, ch. 1

Main point

Bias can enter at every step: the data we train on, how we build the model, and how people use it.

ML trained on biased data will often give biased predictions.

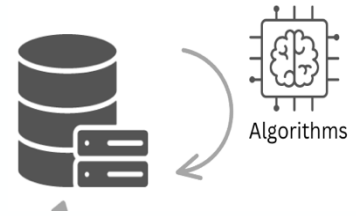


“The biggest threat from artificial intelligence systems is not that they will become smarter than humans, but that they will hard-code sexism, racism, and other forms of discrimination into the digital infrastructure of our societies.” (*Kate Crawford*)

3a. How does machine learning affect DTD?

From the point of view of data quality and data analysis

ML is used in the collection, processing and interpretation of DTD



Main uses of ML in Digital Trace Data: **Recommendation algorithms**

A screenshot of a news feed interface. On the left, there is a video player showing Julian Assange with the ABC News logo and the headline "Julian Assange makes 1st public appearance since his release from prison". To the right of the video, there are three article snippets: one from The Guardian, one from Al Jazeera English, and one from Democracy Now!, all reporting on Julian Assange's public appearance. A "Full Coverage" button is visible at the bottom right of the news feed.

Data processing (interpretation and augmentation)

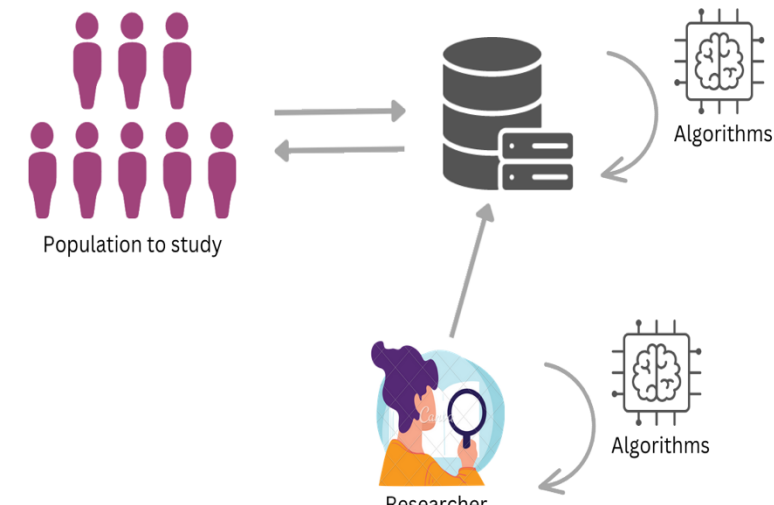
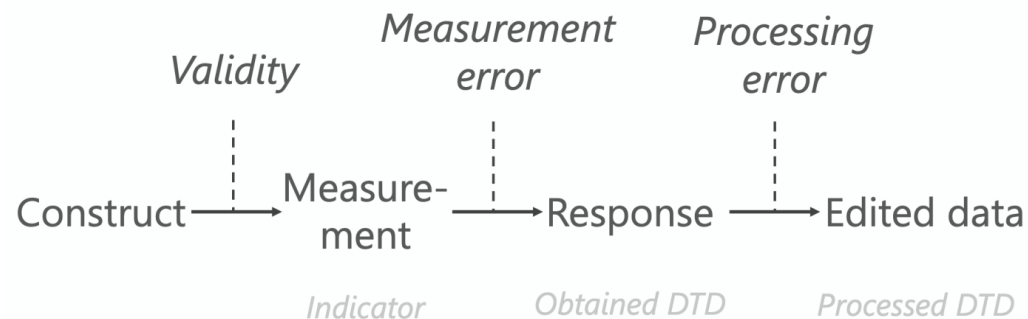
```
],  
"genderInfo" : {  
  "gender" : "male"  
}
```

```
"inferredAgeInfo" : {  
  "age" : [  
    ">50"  
  ],  
  "birthDate" : ""  
}
```

```
{  
  "name" : "Rap",  
  "isDisabled" : false  
},  
{  
  "name" : "Retired life",  
  "isDisabled" : false  
},  
{  
  "name" : "Rom-com films",  
  "isDisabled" : false  
},  
}
```

Errors introduced by ML into DTD

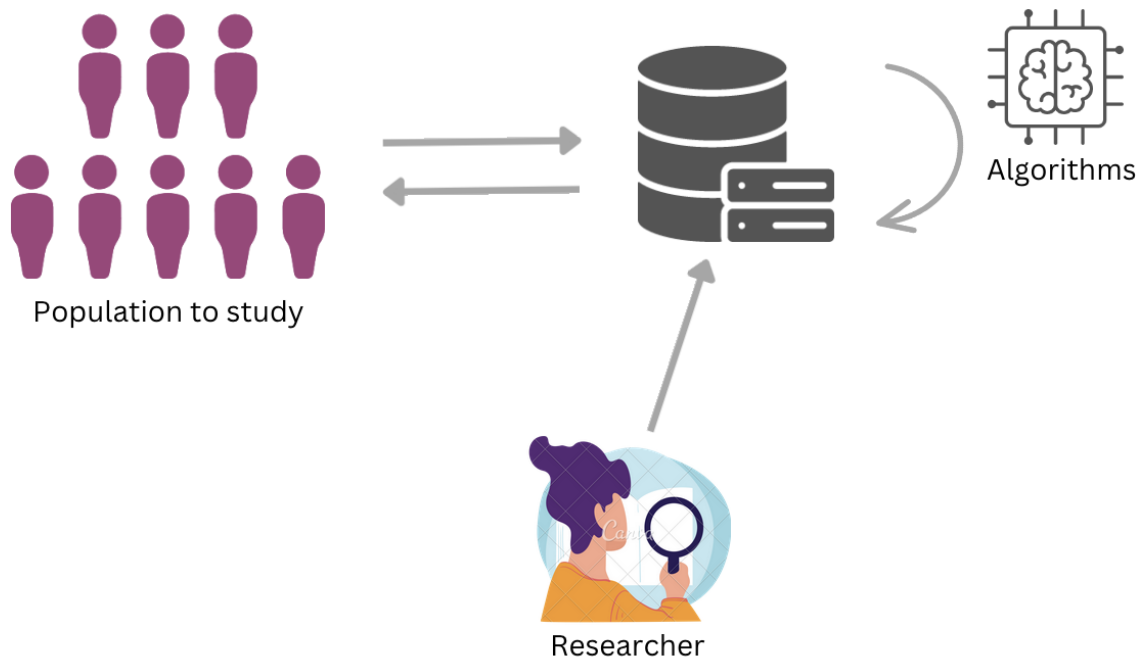
- Total error = measurement error + representation error
- ML impacts mostly measurement-side errors (from lecture 4):
 - On a conceptual level: **validity**
 - Data can be algorithmically confounded (recommendation algorithms)
 - **Measurement error:**
 - When the platform has augmented the data with ML (e.g. inferred gender)
 - **Processing error:**
 - When researchers use ML to process data



Problems with validity

Facebook uses the “clustering coefficient” to recommend friends: e.g., if you have two friends, Sanne and Joep, that are not Facebook friends, Facebook will suggest Sanne and Joep to add each other as friends.

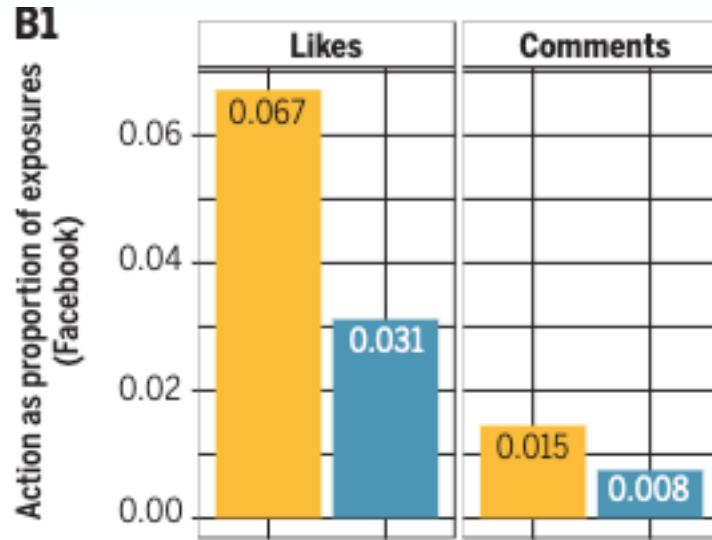
Your measurement of social closure (clustering coefficient) is measuring *both* social closure and the effect of the algorithm.



Problems with validity

More diverse sources!

Algorithmic feed Chronological feed



People less engaged!

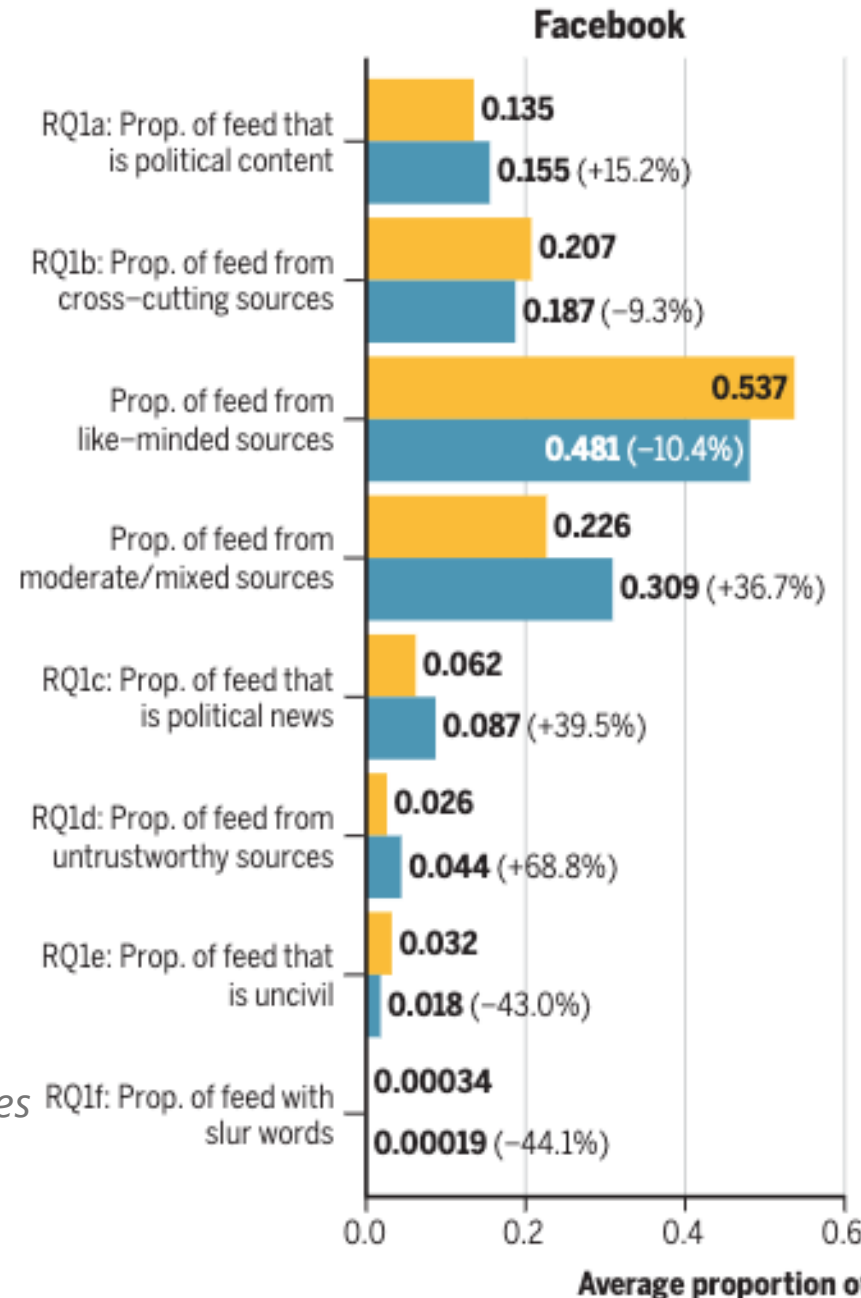
Chronological instead of algorithmic feed (Facebook & Instagram, 3 months, 2020 US election):

- less time on the platforms and less engagement
 - more political and more untrustworthy content
 - no significant change in polarization or other key attitudes
- **What we observe on platforms is partly the algorithm.**

How do social media feed algorithms affect attitudes and behavior in an election campaign?

Guess et al., 2023, Science

<https://doi.org/10.1126/science.abp9364>



In your group project

Think about how ML may be affecting:

- The text data you collected (validity)
- The labels you infer from the text (processing error)

3b. How does machine learning affect societies?

When machine learning models are being used to make decisions, they cannot be separated from the social and ethical context in which they are applied (*Kit T. Rodolfa, Pedro Saleiro, and Rayid Ghani, Big Data book*)

- a) Through errors in algorithms impacting people
- b) Through feedback loops
- c) By reinforcing power structures

A) Algorithms make high-impact decisions

ML is used in many crucial areas for human wellbeing:

- Who to hire – CV screening
- Who to promote – performance reviews
- Who to insure – US life insurance premium based on social media
- Who to jail – predictive policing
- Who to kill – “we kill people based on metadata” (US), Gospel/Lavender systems in Gaza.

Remember there are people behind the data: what are the costs of those failures?

Those algorithms are often (1) “opaque,” (2) “beyond dispute or appeal,” and (3) disproportionately impact the underprivileged (Cathy O’Neil)



Cathy O’Neil

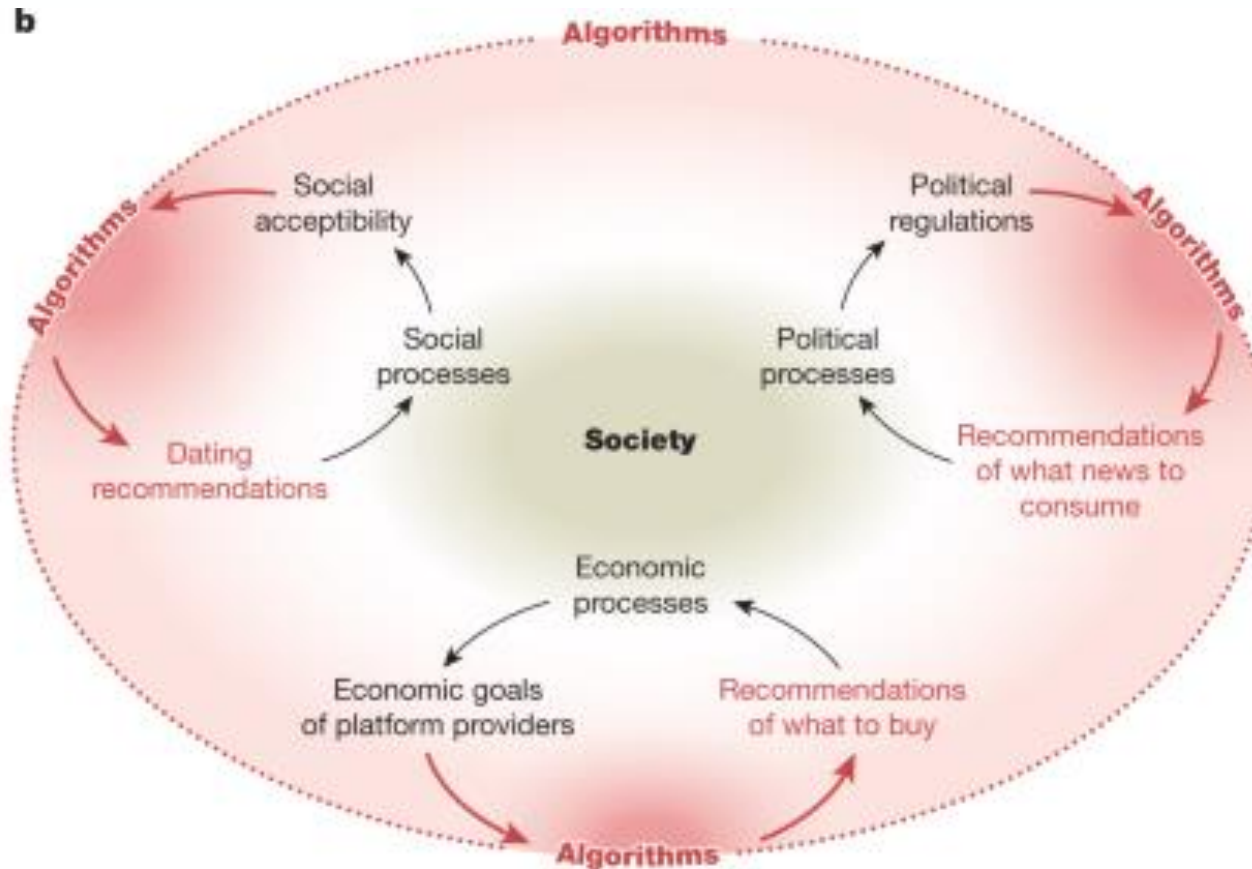
Mathematician (Harvard/MIT/Barnard College)
Worked four years in finance and advertisement

B) Algorithms create feedback effects

We often cannot separate ML from societies

Measuring algorithmically infused societies

Wagner et al, 2021, Nature



the facebook files

How Facebook Hobbled Mark Zuckerberg's Bid to Get America Vaccinated

Company documents show antivaccine activists undermined the CEO's ambition to support the rollout by flooding the site and using Facebook's own tools to sow doubt about the Covid-19 vaccine

Across these samples, anti-vax attitudes appeared in a majority of comments and in visibility positions in comment threads. This is alarming for at least two reasons.

First, the majority of people in the United States and the United Kingdom hold pro-vaccine attitudes and intend on getting a vaccine, suggesting that anti-vax sentiment is over-

- Vaccine hesitancy in comments is rampant.
- Our ability to detect vaccine-hesitant comments is back non-existent elsewhere.

We know that COVID vaccine hesitancy has the potential to cause us to believe that this compounds the necessity of deploying temporary risking over-enforcement. This will minimize harm while we develop approaches, and reduce the risk of problem escalation.

Facebook told the White House to focus on the ‘facts’ about vaccine misinformation. Internal documents show it wasn’t sharing key data.

The tech giant meticulously tracked how misleading medical information spread — but didn’t tell policymakers, even as they demanded it do so.

<https://www.washingtonpost.com/technology/2021/10/28/facebook-covid-misinformation>

https://www.wsj.com/articles/facebook-mark-zuckerberg-vaccinated-11631880296?mod=article_inline

Upvoting extremism: Collective identity formation and the extreme right on Reddit

[Tiana Gaudette](#)  , [Ryan Scrivens](#) , [...], and [Richard Frank](#)  [View all authors and affiliations](#)

[Volume 23, Issue 12](#) | <https://doi.org/10.1177/1461444820958123>

 Contents

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Abstract

Since the advent of the Internet, right-wing extremists and those who subscribe to extreme right views have exploited online platforms to build a collective identity among the like-minded. Research in this area has largely focused on extremists' use of websites, forums, and mainstream social media sites, but overlooked in this research has been an exploration of the popular social news aggregation site Reddit. The current study explores the role of Reddit's unique voting algorithm in facilitating "othering" discourse and, by extension, collective identity formation among members of a notoriously hateful subreddit community, r/The_Donald. The results of the thematic analysis indicate that those who post extreme-right content on r/The_Donald use Reddit's voting algorithm as a tool to mobilize like-minded members by promoting extreme discourses against two prominent out-groups: Muslims and the Left. Overall, r/The_Donald's "sense of community" facilitates identity work among its members by creating an environment wherein extreme right views are continuously validated.

Simil



C) ML reinforces power structures

"Algorithms are opinions embedded in code."

Cathy O'Neil, TED 2017

A model needs past data, a definition of success, and an application.
Someone chooses them.

Think – pair – share

In 2009, DC Public Schools introduced IMPACT, a system for rating teachers. The model predicts how much each student's test score should improve during the year, then credits or blames the teacher for the gap between that prediction and the actual score.

Sarah Wysocki, a 5th-grade teacher at MacFarland Middle School, got excellent reviews from her principal and from parents, but a low value-added score. In 2011 she was fired, along with 205 other DC teachers.

- 0. Was the algorithm biased?
- 1. Who built it, and who decided to use it?
- 2. Whose goals does it serve?
- 3. Who benefits, and who is harmed? Can they appeal?

1 min alone · 3 min in pairs



Examining power: data feminism

Data is never neutral they are always the biased output of unequal social, historical, and economic conditions

Data feminism = who has power, who doesn't, how power shapes data collection, interpretation, and use.

Who is doing the work of data science (and who is not)?

Privilege hazard: those with privilege often don't see how data systems disadvantage others.
e.g. Face recognition not detecting black people.

Whose goals are prioritized in data science (and whose are not)?

Why do we collect data on how to predict crime, instead of how to support those in need?
e.g., crash test dummies designed in the size and shape of men: women had a 47 percent higher chance of car injury.

And who benefits from data science (and who is either overlooked or actively harmed)?

Algorithms often surveil the poor more than the rich (Allegheny County child abuse risk scores).

C) ML reinforces power structures



Meredith Whittaker

- Employed by Google for 13 years
- Research Professor at New York University
- Co-founder and faculty director of the AI Now Institute.
- President of Signal

*"Private computational systems marketed as artificial intelligence (AI) are threading through our public life and institutions, **concentrating industrial power, compounding marginalization, and quietly shaping access to resources and information**" (in *The Steep Cost of Capture*, 2021)*

"The commoditization of our data enables an asymmetric redistribution of power that is weighted toward the actors who have access and the capability to make sense of information" (Sarah Myers West, 2017)

Even if the algorithms are unbiased!

Concentration of control

Market share of cloud computing providers

Amazon Web Services Microsoft Azure Google Cloud Other

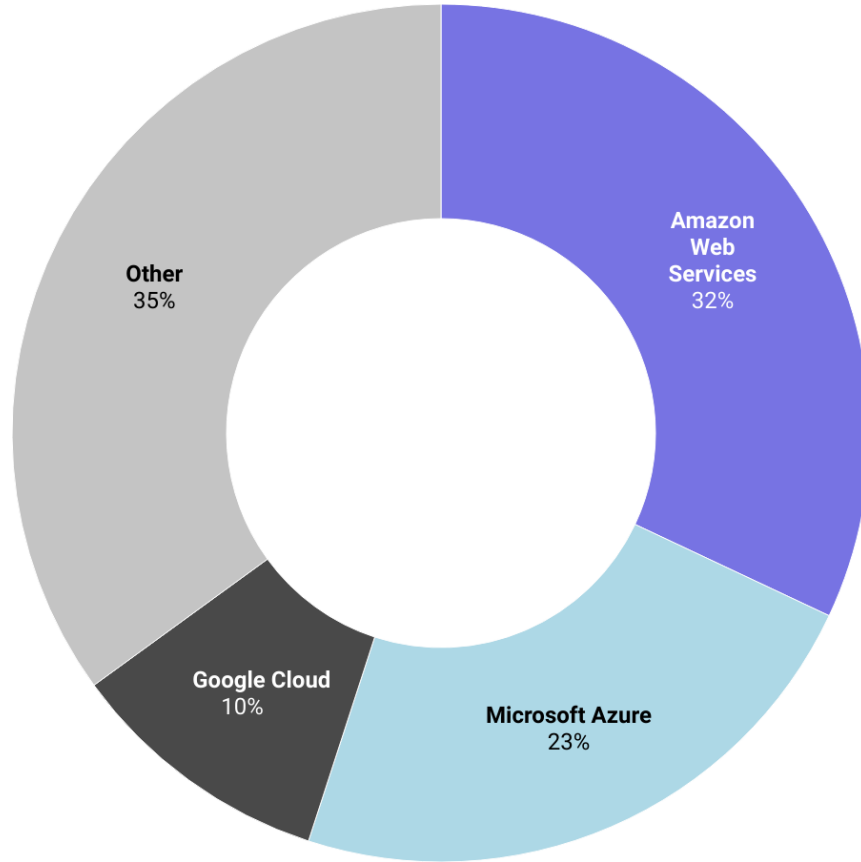


Chart: Exponential View • Source: CRN, AI Now Institute • Created with Datawrapper

Only hegemonic companies have the capital and power to thrive in the new era, reinforcing their power and dominance.

Revenue in 2025:

Amazon:	\$717 bn
<i>Netherlands:</i>	\$489 bn (€433 bn)
Apple:	\$416 bn
Alphabet:	\$403 bn
Microsoft:	\$282 bn
Meta:	\$201 bn

(updated from Babic et al., 2017)



Meta reportedly offered top AI researchers packages of up to \$100M (2025).

We understand social media platforms as ways to share and see content.

Private companies see them as a way to make money, and they influence societal outcomes by controlling information flows and targeting ads and services.

Harm to society is often accepted as part of the business model

THE WALL STREET JOURNAL.

the facebook files 

Facebook Knows Instagram Is Toxic for Teen Girls, Company Documents Show

Its own in-depth research shows a significant teen mental-health issue that Facebook plays down in public



For years they had focus groups, online surveys, diary studies - so this was not one chance finding. A 2019 presentation slide said: "We make body-image issues worse for one in three teenage girls" Another slide said teenagers blamed Instagram for increased levels of anxiety and depression Some 13% of UK teenagers and 6% of US users surveyed traced a desire to kill themselves to Instagram

Instagram response: "Based on our research and feedback from experts, we've developed features so people can protect themselves from bullying, we've given everyone the option to hide 'like' counts and we've continued to connect people who may be struggling with local support organisations."

<https://www.bbc.com/news/technology-58570353>

Performing Platform Governance: Facebook and the Stage Management of Data Relations

Karen Huang¹ · P. M. Krafft²

Received: 2 April 2021 / Accepted: 12 February 2024 / Published online: 4 April 2024

Abstract

Controversies surrounding social media platforms have provided opportunities for institutional reflexivity amongst users and regulators on how to understand and govern platforms. Amidst contestation, platform companies have continued to enact projects that draw upon existing modes of privatized governance. We investigate how social media companies have attempted to achieve closure by continuing to set the terms around platform governance. We investigate two projects implemented by Facebook (Meta)—authenticity regulation and privacy controls—in response to the Russian Interference and Cambridge Analytica controversies surrounding the 2016 U.S. Presidential Election. Drawing on Goffman’s metaphor of stage management, we analyze the techniques deployed by Facebook to reinforce a division between what is visible and invisible to the user experience. These platform governance projects propose to act upon *front-stage data relations*: information that users can see from other users—whether that is content that users can see from “bad actors”, or information that other users can see about oneself. At the same time, these projects relegate *back-stage data relations*—information flows between users constituted by recommendation and targeted advertising systems—to invisibility and inaction. As such, Facebook renders the user experience actionable for governance, while foreclosing governance of back-stage data relations central to the economic value of the platform. As social media companies continue to perform platform governance projects following controversies, our paper invites reflection on the politics of these projects. By destabilizing the boundaries drawn by platform companies, we

How does ML affect societies? – Summary

Data and ML are never neutral

- They always carry social, ethical, and political consequences.
- Are often opaque, can marginalize groups, and create feedback loops.

Power dynamics

- Growth of data & ML concentrates power in big tech companies and elites.

Data feminism

- Provides a critical lens: Who has power? Who is excluded? Who benefits, and who is harmed? Who gets to decide?

4. Dealing with bias in ML

What information do we need to understand bias?

Information on *protected attributes* (gender, sex, class, etc)

Information on *true labels* to create the confusion matrix (per attribute) → Often we need to manually label data, but be mindful of your own biases!

A definition of bias/fairness → There is no universally-accepted definition of what it means for a model to be fair. This is not an excuse for ignoring fairness!

Group A		Predicted criminal	Predicted not criminal
Criminal	10	10	
Not criminal	1	100	

Group B		Predicted criminal	Predicted not criminal
Criminal	10	1	
Not criminal	10	100	

Your turn: is this model fair?

Compute for each group:

- Among the innocent, what share is wrongly labelled criminal? $FP / (FP + TN)$
- Among those labelled criminal, what share is innocent? $FP / (FP + TP)$
- Among the criminals, what share is missed? $FN / (FN + TP)$

Group A		Predicted criminal	Predicted not criminal
Criminal	10	10	
Not criminal	1	100	

Group B		Predicted criminal	Predicted not criminal
Criminal	10	1	
Not criminal	10	100	

Your turn: answers

- Among the innocent, wrongly labelled criminal: A $1/101 \approx 1\%$ · B $10/110 \approx 9\%$ → **False Positive Rate (FPR)**
- Among those labelled criminal, innocent: A $1/11 \approx 9\%$ · B $10/20 = 50\%$ → **False Discovery Rate (FDR)**
- Among the criminals, missed: A $10/20 = 50\%$ · B $1/11 \approx 9\%$ → **False Negative Rate (FNR)**

Group A		Predicted criminal	Predicted not criminal
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Not criminal	1	100	

Group B		Predicted criminal	Predicted not criminal
Criminal	10	1	
Not criminal	10	100	

Errors in algorithms

We need to consider the errors of every algorithm:

- How often they fail?
- For whom do they fail? (**bias/fairness**)

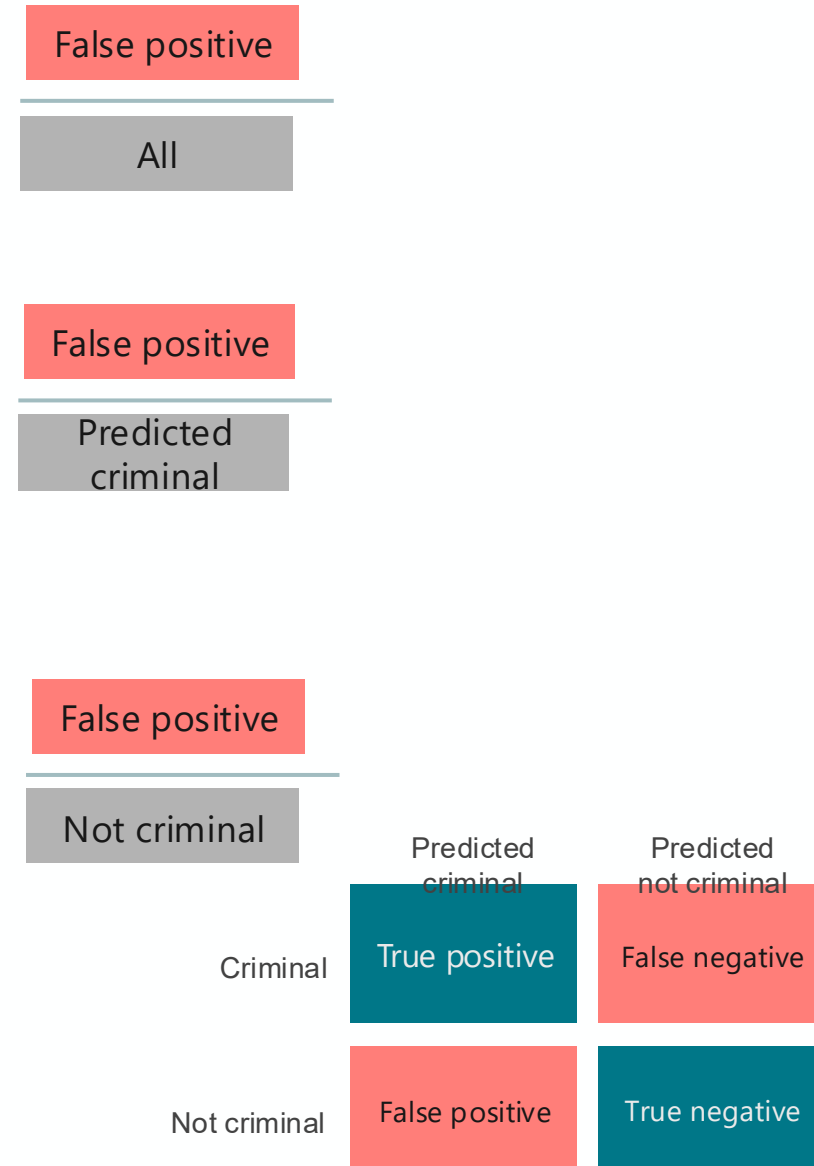
Remember there are people behind the data:

- **What are the costs of those failures?**
 - *Assistive intervention:* "Individuals may be harmed by being incorrectly included in the "low need" population that does not receive an intervention" (**harm = false negatives**, e.g., not supporting somebody in need)
 - *Punitive intervention:* "Individuals may be harmed by being incorrectly included in the "high risk" population that receives an intervention" (**harm = false positives**, e.g. jailing an innocent)

Assistive		Predicted in need	Predicted not in need
In need		True positive 10	False negative 10
Not in need		False positive 1	True negative 100
Punitive		Predicted criminal	Predicted not criminal
Criminal		True positive 10	False negative 10
Not criminal		False positive 1	True negative 100

Different definitions of bias, punitive example

- Among the general population (T), the probability of *being wrongly jailed* is independent of race
- Among the jailed population (those predicted criminals), the probability of being wrongly jailed is independent of race: Parity in **False Discovery Rate**.
→ Focuses on those affected by the intervention
- Among innocents (the actual non-criminals), the probability of being wrongly jailed is independent of race: Parity in **False Positive Rate** (Predictive Equality)
→ Focuses on those who should not be affected by the intervention



No universal definition of fairness!

Different definitions of bias, worked example

Group A

Fixed baseline: each Group B is compared to these rates

Group A	Predicted criminal	Predicted not criminal
Criminal	TP 40	FN 10
Not criminal	FP 10	TN 40

50 criminal, 50 not criminal

FP / group
10 / 100 **10%**

FDR
10 / (40+10) **20%**

FPR
10 / (10+40) **20%**

Base rates differ: 50% criminal in Group A, 20% in every Group B

1 Equal FP / group size

Denominator: everyone in the group

Group B	Predicted criminal	Predicted not criminal
Criminal	TP 10	FN 10
Not criminal	FP 10	TN 70

20 criminal, 80 not criminal

FP / group
10 / 100 **10%** parity
A: 10%

FDR
10 / (10+10) **50%** no parity
A: 20%

FPR
10 / (10+70) **12.5%** no parity
A: 20%

“Among everyone, who is wrongly jailed?”

2 Equal FDR

Denominator: people predicted criminal

Group B	Predicted criminal	Predicted not criminal
Criminal	TP 16	FN 4
Not criminal	FP 4	TN 76

20 criminal, 80 not criminal

FP / group
4 / 100 **4%** no parity
A: 10%

FDR
4 / (16+4) **20%** parity
A: 20%

FPR
4 / (4+76) **5%** no parity
A: 20%

“Among those jailed, who is innocent?”

3 Equal FPR

Denominator: actual non-criminals

Group B	Predicted criminal	Predicted not criminal
Criminal	TP 16	FN 4
Not criminal	FP 16	TN 64

20 criminal, 80 not criminal

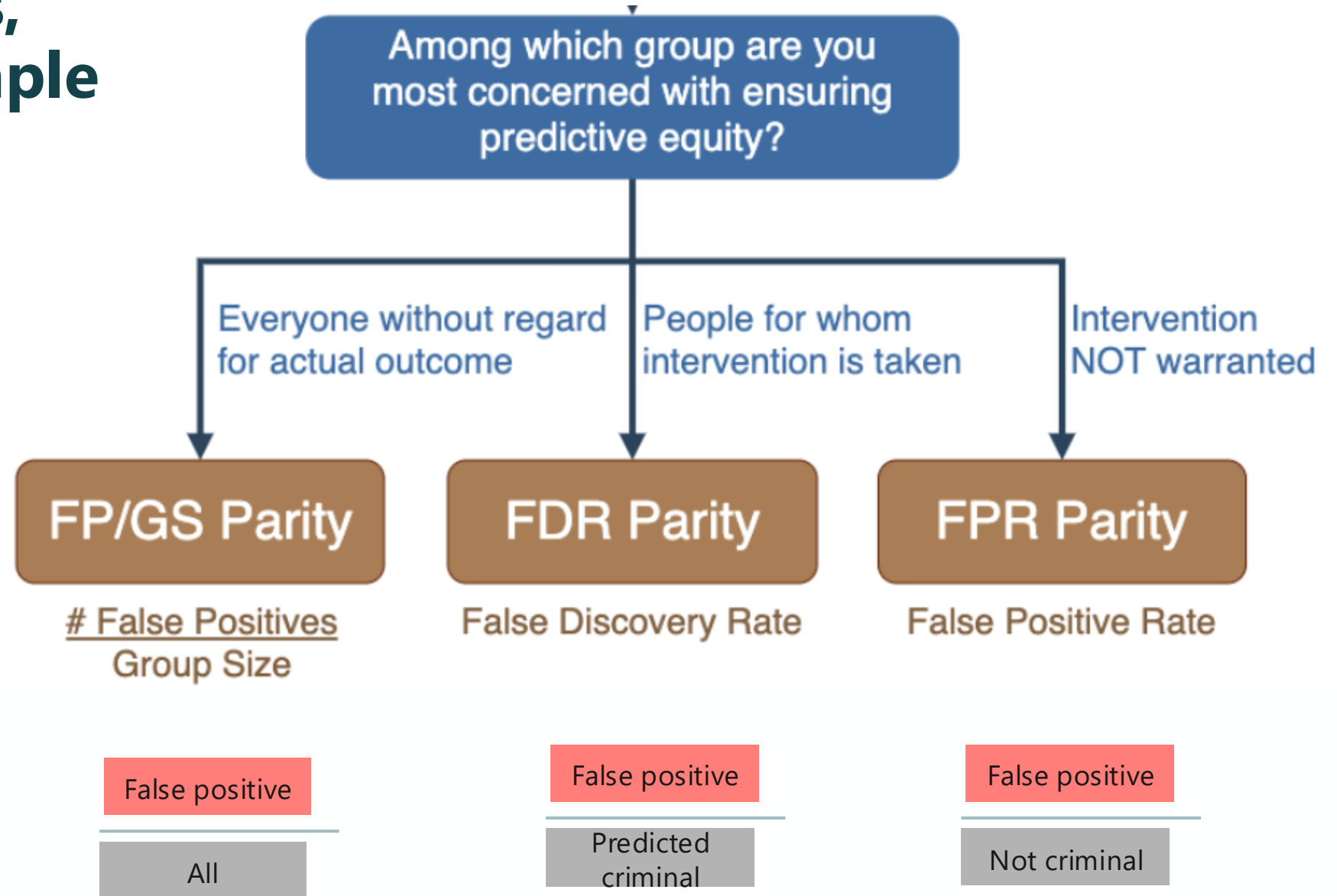
FP / group
16 / 100 **16%** no parity
A: 10%

FDR
16 / (16+16) **50%** no parity
A: 20%

FPR
16 / (16+64) **20%** parity
A: 20%

“Among innocent people, who is jailed?”

Assessing bias, punitive example



Models can be fair and unfair at the same time

	Predicted criminal	Predicted not criminal
Criminal	True positive	False negative
Not criminal	False positive	True negative

Table 11.1: COMPAS Fairness Metrics

Metric	Caucasian	African American
False Positive Rate (<i>FPR</i>)	23%	45%
False Negative Rate (<i>FNR</i>)	48%	28%
False Discovery Rate (<i>FDR</i>)	41%	37%

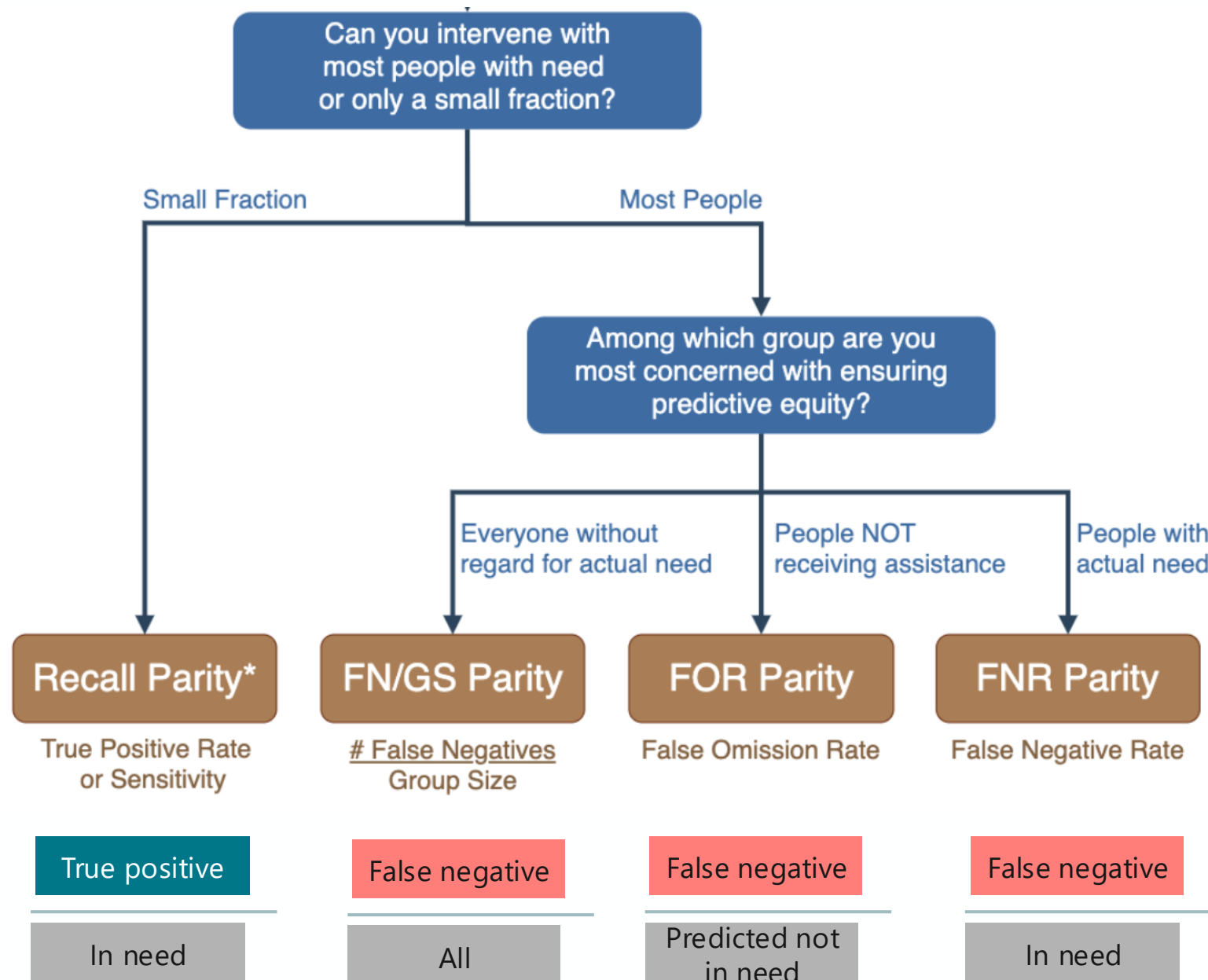
Correctional Offender Management Profiling for Alternative Sanctions (COMPAS): Evaluates the likelihood of an offender committing another crime in the future.

- *FPR*: Among black defendants who did not end up with another arrest, 45% were labeled by the system as high risk, almost twice the rate for whites (23%).
- *FDR*: Among individuals labelled as high risk, a similar fraction of black and white defendants were arrested again.

It's generally impossible for a model to maximize both fairness criteria at the same time

Assessing bias, assistive example

	Predicted in need	Predicted not in need
In need	True positive	False negative
Not in need	False positive	True negative



Dealing with bias in ML

Audit the model to understand bias

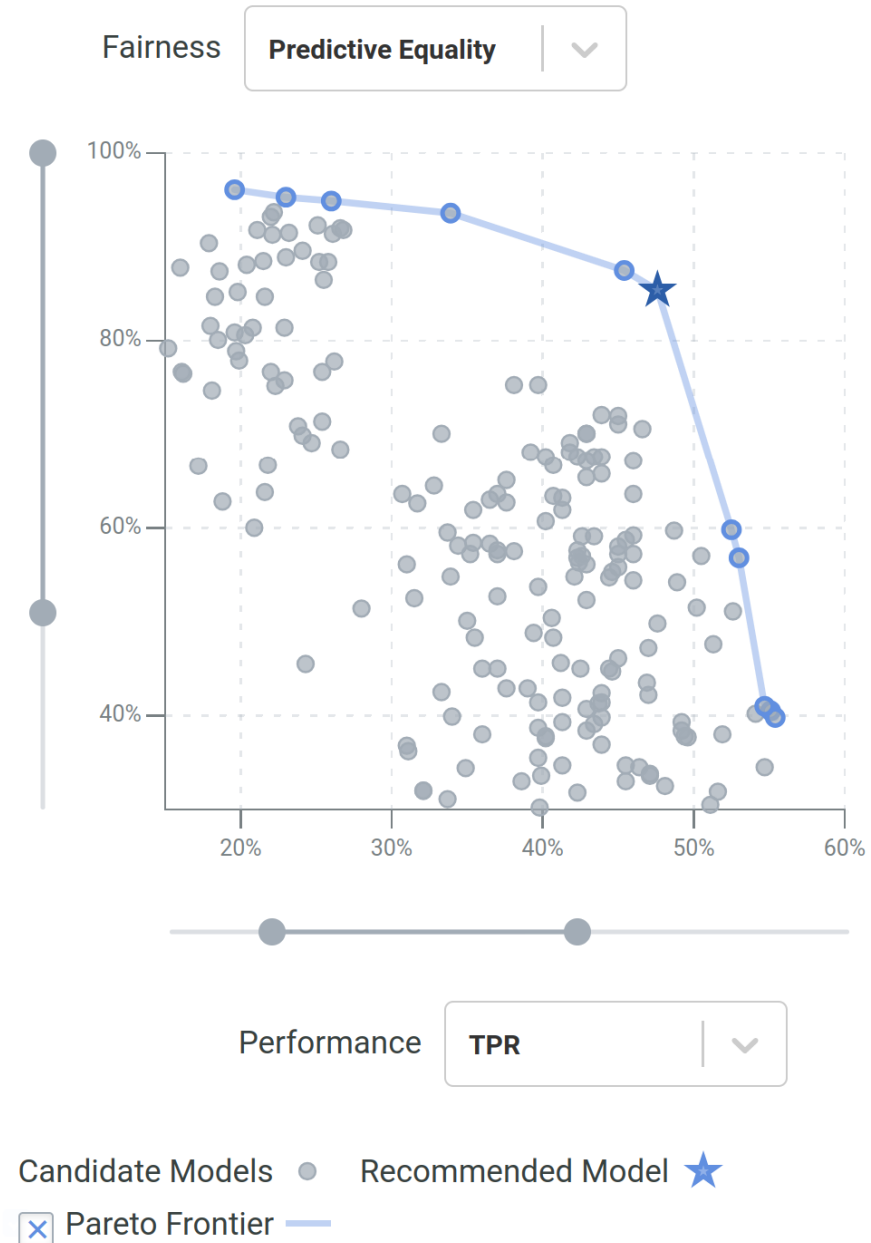
Mitigate bias:

- Removing “protected attributes” (gender/race) generally does *not* mitigate bias. Other features are also correlated with those.
- Test different models and select one with strong performance across fairness and accuracy (Pareto optimality).
- Adjust thresholds to increase/decrease FP or FN. Example: Offer a subsidy to Group A if the model predicts a need with over 50% probability, and to Group B if the need is predicted with over 25% probability.

Consider Intersectionality: Optimizing for one attribute (e.g., gender) may introduce bias in another (e.g., class).

Regularly Test for Bias: Monitor for concept drift and ensure ongoing fairness.

Consider if Bias May Be Acceptable: For example, if the intervention is most useful to a specific subpopulation.



Auditing an algorithm: five questions

- 1. Data integrity:** who is in the data, and what do X and Y really measure? (sample, feature, label bias)
- 2. Definition of success:** who chose it, and does it mean the same for everyone? (label bias)
- 3. Accuracy:** how often and for whom does it fail, and at what cost? (FPR, FDR, FNR)
- 4. Long-term effects:** does it create feedback loops?
- 5. Power:** who decides, who benefits, who is harmed, and can they appeal?

Adapted from Cathy O'Neil's four audit steps (TED, 2017), with the "who" questions of Data Feminism (ch. 1)

TODAY

Lecture

1. What is machine learning and how can we use it?
 - Explain machine learning in your own words
2. Why are ML models biased? Five sources of bias
 - Distinguish the five sources of bias in an ML model
3. How ML affects digital trace data and societies
 - Explain how ML changes what we measure in DTD, and how it can reinforce power
4. Measuring and dealing with bias in ML
 - Compute and compare error rates across groups to assess bias

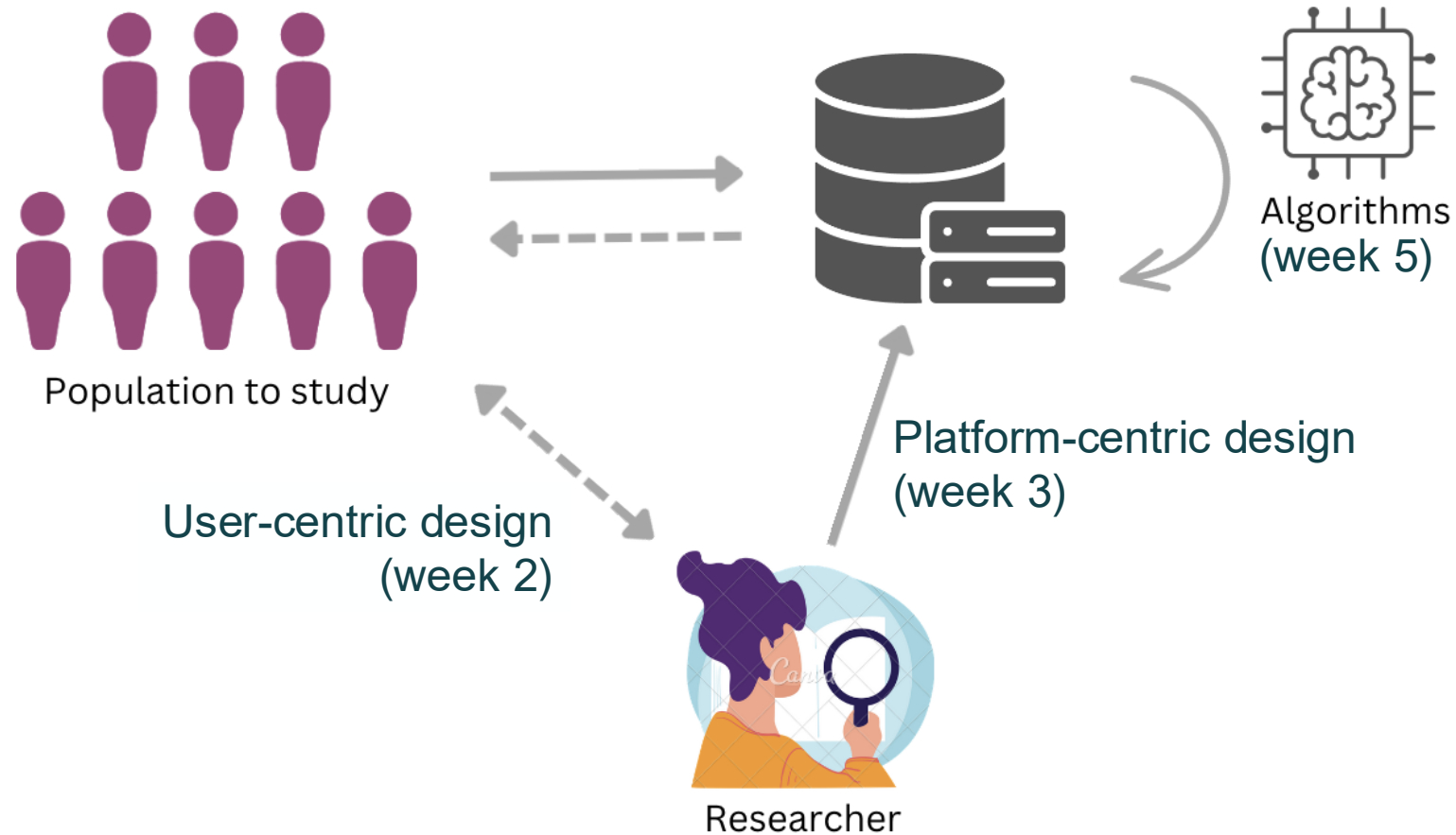
Lab

Discuss power dynamics behind ML models

Apply an ML model to text data

Audit an ML model

Summary of the course



Week 4: Errors in DTD
Week 6: Ethics and Legislation
Week 9: Recap and Q&A